

Start Job Example

Start Job Example – Part 1

Task 1: ^

Server: ||

^ = arrival task

|| = time delimiter

x = execution

o = idle

Task Parameter Table

Tasks C T

Tau_1 1 3

Server Parameter Table

Q_s 1

T_s 3

Server Variable Table

Budget 1

Server

Deadline 3

Utilization ?

Pseudo Code Soft

> New Job Arrival:

If [pending jobs]:

enqueue new job

Else if [$U_s < c_s / (d_s - r)$]:

$c_s = Q_s$

$d_s = r + T_s$

Start job

Budget exhausted:

$c_s = Q_s$ (replenish)

$d_s = d_s + T_s$ (reset deadline)

Start Job Example – Part 2

Task 1: ^x||

Server: ||1||

^ = arrival task

|| = time delimiter

x = execution

o = idle

Task Parameter Table

Tasks C T

Tau_1 1 3

Server Parameter Table

Q_s 1

T_s 3

Server Variable Table

Budget 0

Server
Deadline 3

Utilization ?

Pseudo Code Soft

New Job Arrival:

If [pending jobs]:

enqueue new job

Else if [$U_s < c_s / (d_s - r)$]:

$c_s = Q_s$

$d_s = r + T_s$

> Start job

Budget exhausted:

$c_s = Q_s$ (replenish)

$d_s = d_s + T_s$ (reset deadline)

Start Job Example – Part 3

Task 1: ^x||

Server: ||1||

^ = arrival task

|| = time delimiter

x = execution

o = idle

Task Parameter Table

Tasks C T

Tau_1 1 3

Server Parameter Table

Q_s 1

T_s 3

Server Variable Table

Budget 0

Server

Deadline 3

Utilization ?

Pseudo Code Soft

New Job Arrival:

If [pending jobs]:

enqueue new job

Else if [$U_s < c_s / (d_s - r)$]:

$c_s = Q_s$

$d_s = r + T_s$

Start job

> Budget exhausted:

$c_s = Q_s$ (replenish)

$d_s = d_s + T_s$ (reset deadline)

Start Job Example – Part 4

Task 1: ^x||

Server: ||1||

^ = arrival task

|| = time delimiter

x = execution

o = idle

Task Parameter Table

Tasks C T

Tau_1 1 3

Server Parameter Table

Q_s 1

T_s 3

Server Variable Table

Budget 1

Server

Deadline 3

Utilization ?

Pseudo Code Soft

New Job Arrival:

If [pending jobs]:

enqueue new job

Else if [$U_s < c_s / (d_s - r)$]:

$c_s = Q_s$

$d_s = r + T_s$

Start job

Budget exhausted:

$> c_s = Q_s$ (replenish)

$d_s = d_s + T_s$ (reset deadline)

Start Job Example – Part 5

Task 1: ^x||

Server: ||1||

^ = arrival task

|| = time delimiter

x = execution

o = idle

Task Parameter Table

Tasks C T

Tau_1 1 3

Server Parameter Table

Q_s 1

T_s 3

Server Variable Table

Budget 1

Server

Deadline 6

Utilization ?

Pseudo Code Soft

New Job Arrival:

If [pending jobs]:

enqueue new job

Else if [$U_s < c_s / (d_s - r)$]:

$c_s = Q_s$

$d_s = r + T_s$

Start job

Budget exhausted:

$c_s = Q_s$ (replenish)

$\geq d_s = d_s + T_s$ (reset deadline)

Start Job Example – Part 6

Task 1: ^x||o||

Server: ||1||1||

(server immediately replenished to prevent reaching 0)

^ = arrival task

|| = time delimiter

x = execution

o = idle

Task Parameter Table

Tasks C T

Tau_1 1 3

Server Parameter Table

Q_s 1

T_s 3

Server Variable Table

Budget 1

Server

Deadline 6

Utilization ?

Pseudo Code Soft

New Job Arrival:

If [pending jobs]:

enqueue new job

Else if [$U_s < c_s / (d_s - r)$]:

$c_s = Q_s$

$d_s = r + T_s$

Start job

Budget exhausted:

$c_s = Q_s$ (replenish)

$d_s = d_s + T_s$ (reset deadline)

Start Job Example – Part 7

Task 1: ^x||o||o^
Server: ||1||1||1^

^ = arrival task time delimiter
|| = time delimiter
x = execution
o = idle

Task Parameter Table

Tasks C T
Tau_1 1 3

Server Parameter Table

Q_s 1
T_s 3

Server Variable Table

Budget 1
Server
Deadline 6
Utilization ?

Pseudo Code Soft

New Job Arrival:

 If [pending jobs]:

 enqueue new job

 Else if [$U_s < c_s / (d_s - r)$]:

$c_s = Q_s$

$d_s = r + T_s$

 Start job

Budget exhausted:

$c_s = Q_s$ (replenish)

$d_s = d_s + T_s$ (reset deadline)